The rank-two 4×4 partial-isometry conjecture is answered affirmatively in the literature

Literature-status note for run `fa_banach_001`

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Status: literature already answered. Garcia–Wogen prove the stronger statement that every partial isometry of rank at most two is complex symmetric, and also that every partial isometry in dimension at most four is complex symmetric.

Source conjecture

Conjecture 1 on PDF page 9 of Tener's *Unitary equivalence to a complex symmetric matrix: an algorithm* [1] states:

> Every rank-two 4×4 partial isometry is unitarily equivalent to a complex symmetric matrix.

Decisive literature answer

Garcia and Wogen's separate paper *Complex symmetric partial isometries* [2] explicitly discusses Tener's numerical evidence for this exact four-dimensional assertion. On PDF page 2, their Theorem 2 characterizes complex symmetric partial isometries via compression to the initial space, and they immediately derive:

> **Corollary 1.** Every partial isometry of rank ≤ 2 is complex symmetric.
>
> **Corollary 2.** Every partial isometry on a Hilbert space of dimension ≤ 4 is complex symmetric.

Both corollaries contain Tener's rank-two 4×4 case. In finite dimension, an operator is complex symmetric exactly when some orthonormal basis gives it a complex symmetric matrix, so this is the source's UECSM property.

Provenance, chronology, and scope

The supporting arXiv posting is dated 26 July 2009, shortly before Tener's 15 August 2009 arXiv posting, but Garcia–Wogen cite Tener's earlier manuscript as reference [13] and explicitly identify the same assertion. Thus this is an explicitly linked independent/separate-paper answer, not a same-paper ask-and-answer passage. The answer is affirmative and complete; no part of Conjecture 1 remains unresolved.

Exact-title, arXiv-id, "rank-two 4×4 partial isometry," "UECSM," and complex-symmetric partial-isometry searches found this decisive match. The local run indexes contained no previous packet for either arXiv id.

References

[1] J. E. Tener, *Unitary equivalence to a complex symmetric matrix: an algorithm*, J. Math. Anal. Appl. 341 (2008), 640–648; arXiv:0908.2201.

[2] S. R. Garcia and W. R. Wogen, *Complex symmetric partial isometries*, J. Funct. Anal. 257 (2009), 1251–1260; arXiv:0907.4486.